AWS Storage Services — Practical Assignments (Beginner)

Module: Cloud Storage Fundamentals (Amazon EBS · Amazon S3 · Amazon EFS) **Format:** Scenario-based hands-on labs **Level:** S3 and EFS are beginner; EBS is medium (extra steps: resize, snapshot, copy) **Prerequisite:** A free AWS account

> **How to use these:** Each assignment gives you a short client scenario and tasks to do in the AWS Console. Submit screenshots of each step plus short written answers. Keep it practical — no databases, no production setup.

Assignment 1 — Amazon S3

Scenario: "Host a Simple Website Without a Server"

A client runs a small bakery called **Sweet Crumbs**. They have a one-page website (an `index.html`, a CSS file, and one logo image). They do **not** want to pay for or manage an EC2 server. They ask you: *"Can you host my website using only storage, no server?"*

Tasks

1. Create an **S3 bucket** for the website. (Note what happens if your bucket name is already taken — and explain the bucket naming rule from the module.)
2. Upload the 3 files: `index.html`, a CSS file, and a logo image.
3. Turn on **Static Website Hosting** for the bucket.
4. Open the website URL S3 gives you and confirm the page loads. Share the URL.

Short Questions

5. When you first created the bucket, was it public or private by default? (Answer using the fact from the module.)
6. The client asks, *"Do I need an EC2 server to run this site?"* — answer yes or no and explain in one line.
7. S3 stores data as ______ inside ______. Fill in the two blanks from the module.

Submit: Website URL + screenshots of the bucket and uploaded files + answers to Q5–Q7.

Assignment 2 — Amazon EBS *(Medium Level)*

Scenario: "Add Storage, Back It Up, and Make a Copy for a Teammate"

A client has launched **one EC2 instance** for a small project. They want three things: (1) extra disk space to store their files, (2) a **backup** so they don't lose their work, and (3) a **copy of that storage** they can hand to a teammate to experiment with. They also keep running out of space and ask if they can grow the disk *without* shutting the server down.

Part A — Add Storage

1. Launch one EC2 instance. Find the **EBS volume** that was automatically created with it (the root volume). Note its type and what is stored on it.
2. Create a **new EBS volume** using the general-purpose **gp3** type (the standard choice from the module). When creating it, make sure it is in the **same Availability Zone** as your instance — and explain in your submission why this is required.
3. **Attach** the new volume, then mount it on the instance and create a test file (e.g., `myfile.txt`).
4. **Stop and start** the instance. Confirm your file is still there.

Part B — Resize, Backup & Copy

5. The volume is filling up. **Resize** (increase) the volume to a larger size **without stopping the instance.** State what the module claims about resizing volumes.
6. Create an **EBS Snapshot** of your volume. In your submission, state **where snapshots are stored** according to the module — and whether you can see them in that service's console.
7. From your snapshot, **create a new volume** and attach it to the instance. This is the "copy" for the teammate. Confirm `myfile.txt` exists on the copy too.

Short Questions

8. Why did your file survive even after stopping the instance? (Use the word the module uses for this property.)
9. A teammate in a **different Availability Zone** wants to attach your original volume directly. Explain why that won't work, citing the EBS rule from the module.
10. The client now wants **two EC2 instances to share the exact same volume at the same time**. Is this possible with EBS? If yes, which volume types and which feature make it possible? If no, which service should they use instead?
11. How does EBS billing work — do you pay for storage **provisioned** or storage **used**? Give the approximate gp3 price per GB from the module.

Submit: Screenshots of (a) the attached volume, (b) the file still present after restart, (c) the resize, (d) the snapshot, and (e) the file present on the new copied volume — plus written answers to Q8–Q11.

Assignment 3 — Amazon EFS

Scenario: "Two Servers That Share the Same Files"

A client has **two small EC2 instances** (both Linux). Right now each instance has its own separate storage, so a file saved on one does **not** appear on the other. They ask: *"Can both servers share the same folder, so when I save a file on one, the other sees it too?"*

Tasks

1. Create an **Amazon EFS** file system.
2. **Mount** the EFS file system on **both** EC2 instances.
3. On Instance 1, create a file (e.g., `hello.txt`) inside the EFS folder.
4. On Instance 2, check the same EFS folder and confirm `hello.txt` appears. Screenshot both.

Short Questions

5. What did EFS let you do that a single EBS volume could not? (One line.)
6. The client also has a **Windows** server and wants to mount EFS on it too. Can they? Explain the constraint from the module.
7. The module compares EFS to an everyday thing — what is it? ("EFS is like a ______ that many computers use at once.")

Submit: Screenshots showing the same file on both instances + answers to Q5–Q7.

Quick Recap (for students)

- **S3** → store files/objects, host a simple website, no server needed.
- **EBS** → a virtual hard drive for **one** EC2 instance; data stays even after stopping.
- **EFS** → a shared folder that **many** Linux EC2 instances can use at the same time.